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THE TERM STRUCTURE OF FORWARD EXCHANGE PREMIUMS AND THE FORECASTABILITY OF SPOT EXCHANGE RATES: CORRECTING THE ERRORS

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Abstract—We develop a framework to extract information regarding subsequent spot rate movements from the term structure of forward exchange premiums while admitting possible deviations from rationality and the presence of risk premiums. Using weekly dollar–sterling, dollar–mark, and dollar–yen data, the restrictions implied by our framework are not rejected, and spot and forward exchange rates together are well represented by a vector error correction model (VECM). Dynamic out-of-sample forecasts up to one year ahead indicate that the VECM is strikingly superior to a range of alternative forecasts, including a random walk and standard spot-forward regressions.

I. Introduction

THIS paper revisits one of the most persistent questions in international finance: does the forward exchange rate contain useful information about the future path of the spot exchange rate? In the literature this question has been largely bound up with the question of market efficiency, since under the simple (risk-neutral) efficient markets hypothesis, the forward rate is the optimum predictor of the future spot rate. There is now overwhelming evidence supporting rejection of this simple form of the efficient markets hypothesis. In this paper we develop an empirical framework that is able to accommodate rejection of the simple efficiency hypothesis while still allowing forward premiums to contain information pertinent to future spot rate changes, and which directly implies that the appropriate way to extract this information is through the estimation of vector error correction models (VECMs) in spot and forward rates, rather than through single-equation methods. This approach is general enough to be capable of encompassing both risk aversion and departures from rational expectations on the part of agents, yet is well specified enough to imply testable restrictions on the vector error correction representation. We demonstrate moreover that the information content of the term structure of forward premiums is in fact considerable. Using weekly data on the spot dollar exchange rate and 1-, 3-, 6-, and 12-month forward dollar exchange rates for Germany, Japan, and the United Kingdom, we find that the information contained in the term structure of forward premiums can be used in dynamic out-of-sample forecasting exercises to reduce the root-mean-square error in forecasting the spot rate by around 40% at the one-year forecast horizon relative to a random walk forecast. The approach we suggest also produces forecasts that are superior, on the basis of root-mean-square error (RMSE) or the mean absolute error (MAE), to those produced by an unrestricted vector autoregression, by using the forward rate alone, or by using the standard regression of the rate of depreciation onto the lagged forward premium.

In the next section we briefly discuss the background to the view that the information content of forward rates for future spot rates is slight, while in section III we set out our empirical framework. In section IV we discuss the data, and in section V we report empirical test and estimation results based on the VECM representation. In section VI we discuss the results of the forecasting exercises. A final section concludes and draws some implications for future research.

II. Information Content of Forward Premiums

There is now a consensus within the profession that the simple, risk-neutral efficient markets hypothesis (RNEMH) has been decisively rejected (see Hodrick (1987), Froot and Thaler (1990), and Taylor (1995) for surveys). The cornerstone parity condition for empirical tests of the RNEMH has been the uncovered interest parity (UIP) condition by which the expected rate of exchange rate depreciation is just equal to the relevant interest rate differential.[1] Combining UIP with the covered interest parity condition—that the interest differential is just equal to the relevant forward exchange rate premium[2]—and the assumption of rational expectations implies that the forward premium should be an optimum predictor of the future exchange rate depreciation. Thus tests of the RNEMH have most often taken the form of projecting

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[1] That is, the expected exchange rate depreciation is equal to the interest differential on financial assets in the relevant currencies with the same maturity and identical risk characteristics (see Isard (1993)).

[2] There is strong support for the covered interest parity condition (see Taylor (1987, 1989, 1993, 1995)).

the rate of depreciation onto the lagged forward premium in regression relationships of the kind

$$s_{t+n} - s_t = \alpha + \beta(f_{n,t} - s_t) + \epsilon_{t+n} \tag{1}$$

where s_t is the logarithm of the spot exchange rate at time t (domestic price of foreign currency) and $f_{n,t}$ is the logarithm of the n-period forward rate at time t.[3] Under the RNEMH, $\beta = 1$ and ϵ_t is a disturbance term equal to the rational expectations forecast error. In fact, estimates of equations of the form (1) for a wide variety of exchange rates and time periods have generally yielded estimates of β which are closer to minus unity than plus unity.[4] Moreover, Bekaert and Hodrick (1993) show that this parameter tends to vary over the recent float for the major currencies against the dollar, having become more negative during the 1980s.

Projection equations such as equation (1) have also typically been unable to account for much of the observed variance in exchange rate changes. In regressions of this kind applied to dollar exchange rates for nine industrialized countries, Fama (1984), for example, reports coefficients of determination R^2 ranging from 0.00 to 0.04. The results of this research program have thus led some researchers to conclude that forward premiums contain little or no information with respect to future exchange rate movements.

It should be noted, however, that in regressions of the form of equation (1), while the forward premium (the difference between the forward rate and the current spot rate) typically explains only a tiny fraction of subsequent exchange rate movements (as measured by R^2) and the estimated slope coefficient is typically of the opposite sign to that suggested by the RNEMH, the estimated slope coefficient is nevertheless often statistically significantly different from zero. Bilson (1981, p. 440), for example, using monthly data for the period of July 1974 to January 1980 in a regression of this form pooled across nine dollar exchange rates, finds an estimated slope coefficient of -0.822 and an R^2 of 0.029; but the t-ratio for the hypothesis that the coefficient is zero is -4.57. Findings of this kind are not uncommon. This suggests that there is important information in the term structure, which can be extracted.

In the next section we develop a framework which, while remaining agnostic with respect to the degree of risk aversion or rationality of agents' expectations, demonstrates how information may be extracted from forward premiums and used in forecasting exercises. In the empirical work reported below, we show that this information content is in fact considerable. Most importantly, we demonstrate this in an out-of-sample forecasting context.

III. Empirical Framework

Our empirical framework draws upon a similar framework employed by Hall et al. (1992) to study the term structure of treasury bill yields. It predicts that in a $(j + 1)$-variable system of j forward rates and one spot exchange rate, there should exist j cointegrating vectors and exactly one common trend, which propels the nonstationary component of each of the j forward rates and the one spot exchange rate. In fact, the empirical framework predicts that a basis which spans the space of cointegrating relationships is just the vector of the j forward exchange rate premiums. The central assumption of this framework, which is testable, is that deviations from the simple efficient markets hypothesis, whether due to risk aversion or to nonrational expectations, are realizations of a stationary stochastic process. This seems a reasonable assumption, which one should expect sensible models of the risk premium or of expectations formation to satisfy. It does, in any case, have testable implications.

Consider a vector y_t comprised of the logarithm of the spot exchange rate s_t and the logarithm of j forward exchange rates at horizons $h(1), \ldots, h(j)$,

$$y_t = [s_t, f_{h(1),t}, f_{h(2),t}, \ldots, f_{h(j),t}]'. \tag{2}$$

We suppose that the spot exchange rate possesses a unit root (a hypothesis which, as we discuss below and in common with the literature, we are unable to reject empirically on our data) and evolves according to

$$s_t = z_t + v_t \tag{3}$$

where v_t is a zero-mean stationary stochastic process and z_t is a unit-root process which, for the purposes of exposition but without loss of generality, we assume to be first order,

$$z_t = \gamma + z_{t-1} + e_t. \tag{4}$$

We wish to show that, so long as the deviations (to be defined presently) from the RNEMH are stationary, then each of the forward rates in the term structure will inherit the stochastic trend z_t. The unobserved components model (equations (3) and (4)) is used for generality and expositional clarity. It should be noted, however, that the framework encompasses the case where s_t is generated by a standard unit-root process of the form $(1 - L)Q(L)s_t = v_t$, where $Q(L)$ is a polynomial in the lag operator L, with the roots of $Q(x) = 0$ outside the unit circle.

Let the mathematical expectation of $s_{t+h(j)}$ based on the information set available at time t, Ω_t, be $E(s_{t+h(j)} \mid \Omega_t)$.

[3] Exchange rates are normally expressed in logarithms in this literature because of the "Siegel paradox" that one cannot have, simultaneously, an unbiased expectation of, say, the future dollar–mark rate (marks per dollar) and the future mark–dollar rate (dollars per mark) since $E(1/s) \neq 1/E(s)$. See Taylor (1995) for a discussion of this issue.

[4] See, for example, Bilson (1981), Longworth (1981), Huang (1984), Gregory and McCurdy (1984), Fama (1984), Cumby and Obstfeld (1984), McCallum (1994). Froot (1990) reports that the average estimated value of β as in equation (1), across 75 published estimates, is -0.88. Although this average should not be given undue weight (neither the studies nor the data are independent, thus there may be biases in the refereeing procedure), it does nevertheless give an informal impression of the direction of the published evidence. Taylor (1995) contains a comprehensive review of this literature.

Then we can define the deviation from the RNEMH at horizon $h(j)$, $\phi_{h(j),t}$, say, as

$$\phi_{h(j),t} \equiv f_{h(j),t} - E(s_{t+h(j)} \mid \Omega_t). \tag{5}$$

Combining equations (3) and (5) we obtain an expression for the forward exchange rate at horizon $h(j)$,

$$f_{h(j),t} = h(j)\gamma + z_t + E(v_{t+j} \mid \Omega_t) + \phi_{h(j),t}. \tag{6}$$

Comparing equations (3) and (6), we see that if $\phi_{h(j),t}$ is a stationary stochastic process, then the forward exchange rate at horizon $h(j)$ and the spot rate share a common stochastic trend z_t and are cointegrated such that the forward premium at horizon $h(j)$ is a stationary stochastic process,

$$f_{h(j),t} - s_t = h(j)\gamma + E(v_{t,j} - v_t \mid \Omega_t) + \phi_{h(j),t}. \tag{7}$$

It follows that, among the j forward rates and the spot exchange rate contained in y_t, there will exist at least j cointegrating vectors that are defined by the j forward premiums $f_{h(1),t} - s_t, f_{h(2),t} - s_t, \ldots, f_{h(j),t} - s_t$ so long as the departures from the RNEMH at all horizons are stationary stochastic processes. However, since equations (3) and (6) imply that all $j + 1$ variables in y_t share a common stochastic trend z_t, we know from the results of Stock and Watson (1988) that there will exist *exactly* j independent cointegrating vectors among the $j + 1$ variables in y_t. Thus this empirical framework has the following empirical implications.

First, a vector comprised of the spot exchange rate and j forward exchange rates should be well represented by a VECM. This follows from equations (3), (6), and (7) and the Granger representation theorem (Engle and Granger (1987)). Second, there should exist a unique common trend and thus exactly j cointegrating vectors in such a system, an implication that follows from equations (3) and (6) and the Stock–Watson common trends representation theorem (Stock and Watson (1988)). Third, a basis for this space of j cointegrating vectors should be defined by the j forward premiums in this system $f_{h(1),t} - s_t, f_{h(2),t} - s_t, \ldots, f_{h(j),t} - s_t$. This follows from equation (7). We also note that the results reported in Phillips (1991) imply that it must be possible to select a triangular representation of the cointegration space of this system such that each of j variables in the system is cointegrated with the one remaining ''right-hand-side'' variable not included among these j variables.

If exchange rate changes are not Granger caused by any other available information—that is to say, agents have no information useful for forecasting Δs_{t+1} beyond the history of that variable, so that $E(\Delta s_{t+1} | \Omega_t) = E(\Delta s_{t+1} | \Delta s_t, \Delta s_{t-1}, \ldots)$—then our empirical framework implies that the term structure of forward premiums should not contain any information which helps to improve a forecast of the spot exchange rate given the history of the spot exchange rate itself. Thus merely establishing that spot and forward exchange rates are cointegrated does not guarantee that the term structure of forward premiums contains useful information about the future path of the spot exchange rate. Conversely, if exchange rate changes *are* Granger caused by available information other than the history of the spot exchange rate, our framework implies that the term structure of forward premiums should contain information that helps to improve a forecast of the spot exchange rate given the history of the spot exchange rate itself. To see this more clearly, rearrange equation (5),

$$f_{h(j),t} - s_t = E\left(\sum_{i=1}^{h(j)} \Delta s_{t+i} \mid \Omega_t\right) + \phi_{h(j),t}. \tag{8}$$

From equation (8) we see that the empirical framework implies that the forward premium is the optimal forecast of the sum of $h(j)$ future values of Δs_t, plus $\phi_{h(j),t}$. If information other than the history of Δs_t is relevant for forecasting Δs_t, then equation (8) implies that this must be imparted into the forward premium. If Δs_t is not Granger caused by other information, then equation (8) implies that the forward premium must be an exact linear combination of current and lagged Δs_t, plus $\phi_{h(j),t}$.[5]

In effect, previous researchers have tested equation (8) under the assumption that $\phi_{h(j),t}$ is constant. The point to note, however, is that while foreign exchange market participants may be risk averse, and while their expectations may not conform exactly to the rational expectations hypothesis, the market mechanism may nevertheless impart a significant degree of information into the forward rate.

Note that we have deliberately avoided use of the term ''risk premium'' to describe $\phi_{h(j),t}$. This is because we remain agnostic as to the causes of $\phi_{h(j),t}$. Relation (5) defines $\phi_{h(j),t}$ as *any* departure from the simple, risk-neutral efficient markets hypothesis. While a number of researchers have interpreted this as due to risk, Froot and Frankel (1989) used survey data to argue that the bias in the premium as a spot rate forecast is not due entirely to risk. Indeed, these authors cannot reject the hypothesis that all of the bias is due to systematic expectational errors.[6] This is not inconsistent with our framework: $\phi_{h(j),t}$ can be interpreted as the deviation of agents' expectations from full market rationality, as well as being due to risk aversion.[7]

[5] This reasoning is similar to the argument developed in Campbell and Shiller (1987).

[6] Hodrick (1992) suggests, however, that the Froot–Frankel results should be interpreted with caution. In particular, Hodrick notes that while Frankel and Froot are unable to reject the hypothesis of a slope coefficient of unity in the regression of the market survey forecast onto the forward premium, the R^2 in that regression is far from perfect, as it should be if the forward premium were the market's expected rate of depreciation and risk factors were insignificant.

[7] Systematic expectational errors might be generated, for example, by the influence of ''chartist'' or ''technical'' analysts on market traders (Goodhart (1988), Frankel and Froot (1990), Curcio and Goodhart (1991), Taylor and Allen (1992), Allen and Taylor (1993), Froot et al. (1992)), and/or because of learning by some traders (Lewis (1989), Cutler et al. (1990)). See Froot and Thaler (1990) for a general discussion of ''quasi-rational'' behavior in the foreign exchange market.

Before moving on to the empirical results, we should comment on an alternative framework which can be employed to interpret the joint behavior of the spot exchange rate and term structure of forward exchange premiums. Our framework imposes the testable restriction that $\phi_{h(j),t}$, the departure from the RNEMH, is a realization of a stationary stochastic process. If instead $\phi_{h(j),t}$ possesses a unit root, we should be able to reject the hypothesis that each of the j forward premiums $f_{h(1),t} - s_t, f_{h(2),t} - s_t, \ldots, f_{h(j),t} - s_t$ is stationary (Evans and Lewis (1994)). To see this point, suppose that

$$\phi_{h(j),t} = \phi(j)x_t + w_{jt} \tag{9}$$

where w_{jt} is a zero-mean stationary stochastic process, and x_t is a random walk. Using equation (6) we see that

$$f_{h(j),t} = h(j)\gamma + z_t + \phi(j)x_t + E(v_{t+j} \mid \Omega_t) + w_{jt}\,. \tag{10}$$

From equation (10) we see that $f_{h(j),t} - s_t$ inherits the unit root present in x_t. Thus, if this alternative interpretation of the data is correct, we should be able to reject the hypothesis that each of the j forward premiums is stationary. Moreover, unless x_t is proportional to z_t, this interpretation also implies that among the $j + 1$ variables in the system, there are two common trends and thus $j - 1$ cointegrating vectors. Thus a finding of $j - 1$ or fewer cointegrating vectors in a system comprised of j forward exchange rates and the spot exchange rate is evidence in favor of this alternative interpretation.

IV. The Data and Some Empirical Preliminaries

We investigate weekly, point-in-time sampled data on spot and 4-, 13-, 26-, and 52-week forward dollar exchange rates for Germany, Japan, and the United Kingdom, obtained from the Harris Bank database maintained by Richard Levich.[8] The sample runs from 1977:1 through 1993:52, although most of the estimations are carried out using data for the period of 1977:1 through 1990:26, so that the last three and a half years can be reserved for out-of-sample forecasting tests. The choice of starting date reflects the view, first expressed by Hansen and Hodrick (1980) in their classic study of the forecastability of excess returns in the foreign exchange market, that during the early years of floating and until the Rambouillet Agreement of February 1976, market participants may very well have believed that a return to fixed parities was imminent. If this was in fact the case, then departures from the RNEMH during these years would have reflected not only a risk premium, but also an extra component incorporating the effect of a return to fixed parities on expected payoffs to foreign exchange speculation. This is of course the classic ''peso problem'' (Rogoff (1977) and Krasker (1980)). See Evans and Lewis (1994) for an extensive discussion of the peso problem and its effect on stochastic trends.

All data were converted to logarithmic form.

Preliminary unit-root (augmented Dickey–Fuller) tests on the data confirmed the findings, reported in many earlier studies, that the hypothesis of a unit root in the stochastic processes governing spot and forward exchange rates cannot be rejected, although the unit-root hypothesis was rejected for all of the forward premium series. Also, investigations based on regression equations of the form (1) in each case led to an easy rejection of the RNEMH, with the estimated slope coefficients in each case significantly negative and significantly different from unity.[9]

V. A Vector Error Correction Model

Based on the empirical framework developed in section III and our preliminary finding that we are unable to reject the hypothesis that the series under study are integrated of order 1, we investigated a dynamic VECM (Engle and Granger (1987) and Johansen (1991)) for the spot dollar exchange rate and the term structure of forward dollar exchange rates for the United Kingdom, Japan, and Germany.[10] Letting $y_t = [s_t, f_{4,t}, f_{13,t}, f_{26,t}, f_{52,t}]'$ denote the $j + 1 = 5 \times 1$ vector of the system's variables for a particular currency, the VECM can be written

$$\begin{aligned}\Delta y_t = \mu + \Gamma_1 \Delta y_{t-1} + \cdots + \Gamma_{k-1}\Delta y_{t-k+1} \\ + \Pi y_{t-k} + \epsilon_t\end{aligned} \tag{11}$$

where Δ is the first-difference operator. If the matrix Π is of full rank, $r = 5$, the VECM reduces to the usual vector autoregression (VAR) in the levels of stationary variables. If Π is the null matrix so that $r = 0$, the VECM represents a VAR in first differences. The VECM differs from the usual VAR in that it allows for the existence of long-run ''equilibrium'' relationships among a system's variables. If the matrix Π is of reduced rank, $r < 5$, it can be factored into the product of two $5 \times r$ matrices α and β such that

$$\Pi = \alpha\beta' \tag{12}$$

where β' is the $r \times 5$ matrix of the system's r cointegrating vectors, and α is the $5 \times r$ matrix of r adjustment coefficients for each of the system's five equations.

Each cointegrating relationship defines a long-run equilibrium to which the system ultimately returns after a shock. The parameters in the α matrix determine the rate at which

[8] It is well known that one cannot follow the precise conventions of delivery on forward contracts from this data set. Hence as a technical matter, the forward premium in this data set would not be equal to the expected rate of depreciation purely because of this slight mismatch. Fortunately, however, Bekaert and Hodrick (1993) argue convincingly that inference is not substantively affected using the Harris data set versus data sampled to match the market conventions.

[9] See Clarida and Taylor (1993) for details. All hypothesis tests were based on a nominal significance level of 5%.

[10] Although he does not employ the same framework as in the present paper, Bekaert (1995) estimates a vector autoregression in the rates of depreciation and forward premiums for dollar–mark, dollar–sterling, and dollar–yen.

TABLE 1.—TESTS OF COINTEGRATING RANK OF $y_t = [s_t, f_{4,t}, f_{13,t}, f_{26,t}, f_{52,t}]'$

	λ-max Statistic	5% Critical Value	Trace Statistic	5% Critical Value
Dollar–sterling				
H_0: $r \leq 4$	3.729	3.762	3.729	3.762
H_0: $r \leq 3$	16.129	14.069	19.858	15.410
Dollar–mark				
H_0: $r \leq 4$	0.196	3.762	0.196	3.762
H_0: $r \leq 3$	16.245	14.069	16.441	15.410
Dollar–yen				
H_0: $r \leq 4$	2.135	3.762	2.135	3.762
H_0: $r \leq 3$	18.117	14.069	18.554	15.410

Note: Sample period is 1977:1 to 1990:26. Critical values are from Osterwald-Lenum (1992, table 1).

each of the system's variables adjusts in response to lagged deviations from the r cointegrating relationships. Stock and Watson (1988) prove that the long-run behavior of a system of n variables with $r < n$ cointegrating relationships is governed by $n - r$ common stochastic trends. Thus a test for the cointegration rank r is also a test for the number of common trends.

Table 1 presents the results of two tests developed by Johansen (1991) to investigate the hypothesis that the number of cointegrating vectors in a system of n variables is less than or equal to r. Note that the Stock and Watson results cited above imply that this is also a test of the hypothesis that the number of stochastic trends in the n-variable system is greater than or equal to $n - r$. According to both the trace and the λ-max statistics, we cannot reject for any of the three exchange rates the hypothesis that $r \leq 4$, but we can reject at the 5% level the hypothesis that $r \leq 3$.[11] Thus, for sterling, the deutsche mark, and the yen, these findings are consistent with the predictions of the empirical framework that, in a system comprised of a spot exchange rate and j forward exchange rates, exactly one common trend and j cointegrating relations are needed to account for the dynamic behavior of the system.

Another prediction of the empirical framework is that a basis for the space of cointegrating relationships is defined by the vector of $j = 4$ forward premiums $[f_{4,t} - s_t, f_{13,t} - s_t, f_{26,t} - s_t, f_{52,t} - s_t]'$. A likelihood ratio statistic is employed to test this hypothesis. Conditional on there being four cointegrating vectors in the system, the likelihood ratio test statistic for this hypothesis is distributed as χ^2 (4) under the null. The results of this test are reported in table 2. For none of the exchange rates is it possible to reject the hypothesis that the vector of forward premiums defines a basis for the space of $j = 4$ linearly independent cointegrating relationships implied by the estimated VECMs.

TABLE 2.—TESTS OF THE NULL HYPOTHESIS THAT FOUR LINEARLY INDEPENDENT FORWARD PREMIUMS COMPRISE A BASIS FOR THE COINTEGRATION SPACE

	χ^2 (4)	Marginal Significance Level
Dollar–sterling	2.88	58%
Dollar–mark	5.32	26%
Dollar–yen	7.32	12%

Note: Sample period is 1977:1 to 1990:26. The test is conditional on there being four linearly independent cointegrating vectors.

We conclude from this evidence that the empirical framework outlined in section III is well supported by the data. In particular, for all three exchange rates, the spot exchange rate and the term structure of forward rates are well modeled by a VECM. In each system, exactly one common trend and thus four cointegrating vectors are required to explain the dynamic behavior of spot and forward exchange rates. These four cointegrating relations are, as predicted by the analysis, defined by the vector of four forward premiums for each currency.

We now investigate whether or not the term structure of forward premiums contains incremental predictive content for the time path of the spot exchange rate.

Tables 3, 4, and 5 present full-information maximum-likelihood (FIML) estimates of the five-equation VECMs for the sterling, mark, and yen systems, respectively. Parsimony was achieved in these estimates by first estimating a general first-order VECM and then excluding variables whose estimated coefficients were insignificant at the 5% level. A test statistic for the joint exclusion restrictions on each system was also constructed and is reported in the tables.[12]

Of particular interest are the results for the Δs_t equation reported in the first two columns of the tables. As can be seen in table 3, the spot dollar–sterling exchange rate is not exogenous with respect to lagged information contained in the term structure of forward premiums. Indeed, the lagged 13-, 26-, and 52-week forward premiums contain statistically significant information about the future path of the dollar–sterling spot exchange rate that is not contained in the lagged change in the spot rate. Similarly, tables 4 and 5 report that the spot dollar–mark and dollar–yen exchange rates are not exogenous with respect to lagged information contained in the term structure of forward premiums. The entire lagged term structure of forward premiums contains statistically significant information about the future path of

[11] These statistics should, however, be interpreted with care, given the subtleties of testing for cointegration. In particular, note that the critical values used for the Johansen test statistics assume a drift in the VECM representation. If in fact the VECM representation is driftless ($\mu = 0$ in equation (11)), then the appropriate critical values to use are those given in Osterwald-Lenum (1992, table 1*). Based on the λ-max statistic, our inferences would be unchanged using these more conservative critical values—we would still be unable to reject for any of the three exchange rates the hypothesis that $r \leq 4$, but we would reject at the 5% level the hypothesis that $r \leq 3$. Using the trace statistic, however, the same inference could be drawn for the dollar–sterling and dollar–yen exchange rates only at the 10% nominal significance level, and only at around the 15% level for dollar–mark. Given that our inferences are unchanged using the λ-max statistic and, moreover, that there is no a priori reason to constrain the drift to zero, we remain confident that the inferences drawn in this section with respect to the number of independent cointegrating vectors are correct.

[12] The VECM was restricted to first-order lags of the differenced series mainly because the full information maximum-likelihood algorithm became unstable when higher-order lags were included, due to near collinearity. In terms of the residual diagnostics (tables 3, 4, and 5) and out-of-sample forecasting performance (see below), however, the first-order VECM system seems more than adequate.

TABLE 3.—FIML ERROR CORRECTION MODEL FOR FIVE-VARIABLE SYSTEM: DOLLAR–STERLING

Explanatory Variable	Model for Δs_t		Model for $\Delta f_{4,t}$		Model for $\Delta f_{13,t}$		Model for $\Delta f_{26,t}$		Model for $\Delta f_{52,t}$	
	Coeff.	SE	Coeff.	SE	Coeff.	SE	Coeff.	SE	Coeff.	SE
Δs_{t-1}	−1.701	1.026	−1.749	1.023	−2.044	1.023	−2.374	1.034	−3.295	1.074
$\Delta f_{4,t-1}$	1.847	1.030	1.917	1.027	2.432	1.029	2.827	1.042	3.472	1.078
$\Delta f_{13,t-1}$	—	—	−0.058	0.031	−0.407	0.061	−0.467	0.069	—	—
$\Delta f_{26,t-1}$	−0.153	0.041	−0.130	0.032	—	—	—	—	—	—
$\Delta f_{52,t-1}$	−0.011	0.007	—	—	—	—	—	—	−0.189	0.030
$(s - f_4)_{t-1}$	—	—	−0.143	0.045	−0.643	0.115	−0.807	0.193	−0.723	0.324
$(s - f_{13})_{t-1}$	0.203	0.071	−0.921	0.436	−0.981	0.306	0.197	0.072	0.300	0.076
$(s - f_{26})_{t-1}$	−0.178	0.059	0.178	0.059	−0.184	0.093	−0.123	0.060	−0.248	0.062
$(s - f_{52})_{t-1}$	0.521	0.175	0.519	0.174	0.485	0.174	0.339	0.176	−0.734	−0.182
Constant	−0.002	0.001	−0.002	0.001	0.002	0.001	0.003	0.001	0.004	0.001
	$Q(13) = 10.71$ (0.64)		$Q(13) = 14.10$ (0.37)		$Q(13) = 11.32$ (0.58)		$Q(13) = 12.89$ (0.46)		$Q(13) = 12.72$ (0.47)	
				$H(325) = 343.33$ (0.24)	REST(9) = 9.61 (0.38)					

Notes: Sample period is 1977:1 to 1990:26. — indicates that the coefficient was found to be insignificant in the reduction process. The Q-statistics are Ljung–Box statistics computed at 13 autocorrelations of the residual series; H is Hosking's (1980) multivariate portmanteau statistic computed at 13 autocorrelations; REST is a likelihood ratio statistic for the exclusion restrictions. All statistics are distributed as central chi-square under the null hypothesis, with the degrees of freedom indicated. Figures in parentheses are marginal significance levels.

TABLE 4.—FIML ERROR CORRECTION MODEL FOR FIVE-VARIABLE SYSTEM: DOLLAR–MARK

Explanatory Variable	Model for Δs_t		Model for $\Delta f_{4,t}$		Model for $\Delta f_{13,t}$		Model for $\Delta f_{26,t}$		Model for $\Delta f_{52,t}$	
	Coeff.	SE	Coeff.	SE	Coeff.	SE	Coeff.	SE	Coeff.	SE
Δs_{t-1}	−1.789	0.710	−1.734	0.705	−1.876	0.700	−2.284	0.702	−2.344	0.739
$\Delta f_{4,t-1}$	3.267	0.746	3.075	0.736	2.104	0.722	2.589	0.415	2.604	0.740
$\Delta f_{13,t-1}$	−1.799	0.229	−1.6433	0.212	−1.596	0.182	−0.985	0.143	−0.252	0.033
$\Delta f_{26,t-1}$	0.328	0.067	0.310	0.057	0.277	0.040	—	—	—	—
$\Delta f_{52,t-1}$	—	—	—	—	—	—	—	—	—	—
$(s - f_4)_{t-1}$	−0.981	0.213	−0.939	−0.242	0.601	0.171	0.597	0.171	0.596	0.173
$(s - f_{13})_{t-1}$	−0.743	0.202	−0.361	−0.128	−0.261	0.088	−0.283	0.087	−0.167	0.089
$(s - f_{26})_{t-1}$	−0.376	0.143	—	—	−0.224	0.024	—	—	−0.18	0.010
$(s - f_{52})_{t-1}$	−0.061	0.018	0.408	0.135	0.412	0.135	0.366	0.135	−0.712	0.130
Constant	0.006	0.002	0.005	0.002	0.005	0.002	0.006	0.002	0.008	0.002
	$Q(13) = 9.71$ (0.72)		$Q(13) = 8.43$ (0.81)		$Q(13) = 7.65$ (0.87)		$Q(13) = 9.51$ (0.73)		$Q(13) = 10.65$ (0.64)	
				$H(325) = 337.36$ (0.31)	REST(9) = 5.8 (0.76)					

Notes: Sample period is 1977:1 to 1990:26. — indicates that the coefficient was found to be insignificant in the reduction process. The Q-statistics are Ljung–Box statistics computed at 13 autocorrelations of the residual series; H is Hosking's (1980) multivariate portmanteau statistic computed at 13 autocorrelations; REST is a likelihood ratio statistic for the exclusion restrictions. All statistics are distributed as central chi-square under the null hypothesis, with the degrees of freedom indicated. Figures in parentheses are marginal significance levels.

both the dollar–mark and the dollar–yen spot exchange rates that is not contained in the lagged change in the spot rate.

The dollar–mark and dollar–yen estimates are also similar in that all of the forward premiums or error correction terms enter with a negative coefficient. This implies that, when dollar interest rates are high relative to yen and mark rates at any point in the term structure (so that the dollar is trading at a discount against these currencies in the forward market), there is a tendency for the dollar to depreciate. This is qualitatively consistent with long-run uncovered interest rate parity, and is quite different from the ‘‘high-yield strategy’’ which is the typical investment advice implied by a negative estimated coefficient in a regression equation of the form (1). A further similarity between dollar–yen and dollar–mark is that the absolute size of the point estimates of the error correction parameters declines as the maturity of the forward rate rises. This result seems intuitive, since one would expect weekly exchange rate movements to be determined more by developments at the shorter end of the term structure of forward premiums.

The relationship between weekly exchange rate movements and lagged forward premiums for the dollar–sterling exchange rate is, however, far more complex and does not yield such easy intuitive interpretation.

VI. Out-of-Sample Forecasting

As noted above, the finding that forward premiums are statistically significant *in sample* in explaining changes in the spot rate is not entirely novel. In order to assess the usefulness of the information in the term structure of forward exchange rates, therefore, the following out-of-

TABLE 5.—FIML ERROR CORRECTION MODEL FOR FIVE-VARIABLE SYSTEM: DOLLAR–YEN

Explanatory Variable	Model for Δs_t		Model for $\Delta f_{4,t}$		Model for $\Delta f_{13,t}$		Model for $\Delta f_{26,t}$		Model for $\Delta f_{52,t}$	
	Coeff.	SE	Coeff.	SE	Coeff.	SE	Coeff.	SE	Coeff.	SE
Δs_{t-1}	−1.614	0.613	−1.332	0.712	−1.477	0.547	−2.147	0.814	−2.147	0.784
$\Delta f_{4,t-1}$	—	—	3.221	0.824	1.998	0.652	2.554	0.458	2.648	0.812
$\Delta f_{13,t-1}$	−1.512	0.231	2.441	0.826	−1.687	0.278	−0.889	0.210	−0.288	0.112
$\Delta f_{26,t-1}$	−0.331	0.093	0.332	0.062	0.285	0.069	0.604	0.219	—	—
$\Delta f_{52,t-1}$	−0.112	0.053	—	—	—	—	—	—	—	—
$(s - f_4)_{t-1}$	−0.541	0.221	0.331	−0.212	0.801	0.181	0.582	0.211	0.447	0.225
$(s - f_{13})_{t-1}$	−0.527	0.211	−0.355	−0.157	0.641	0.147	0.058	0.019	−0.441	0.159
$(s - f_{26})_{t-1}$	−0.527	0.137	0.847	0.224	−0.147	0.054	0.069	0.018	0.185	0.019
$(s - f_{52})_{t-1}$	−0.083	0.021	0.414	0.118	0.541	0.214	0.381	0.117	−0.841	0.220
Constant	0.008	0.002	0.003	0.001	0.006	0.002	0.008	0.002	0.006	0.002
	$Q(13) = 11.82$ (0.54)		$Q(13) = 10.78$ (0.63)		$Q(13) = 12.13$ (0.52)		$Q(13) = 11.16$ (0.60)		$Q(13) = 10.39$ (0.66)	
				$H(325) = 353.22$ (0.13)	REST(6) = 7.66 (0.26)					

Notes: Sample period is 1977:1 to 1990:26. — indicates that the coefficient was found to be insignificant in the reduction process. The Q-statistics are Ljung–Box statistics computed at 13 autocorrelations of the residual series; H is Hosking's (1980) multivariate portmanteau statistic computed at 13 autocorrelations; REST is a likelihood ratio statistic for the exclusion restrictions. All statistics are distributed as central chi-square under the null hypothesis, with the degrees of freedom indicated. Figures in parentheses are marginal significance levels.

TABLE 6.—RESULTS OF FORECASTING EXERCISES: DOLLAR–STERLING

	VECM (level)	VAR (ratio)	Random Walk (ratio)	Forward Premium Regression (ratio)	Forward Rate (ratio)
Root-mean-square error (RMSE)					
4-week horizon	0.0402	0.959	0.998	0.957	1.002
13-week horizon	0.0707	0.799	0.863	0.774	0.866
26-week horizon	0.0806	0.601	0.689	0.543	0.695
52-week horizon	0.0828	0.445	0.575	0.361	0.579
Mean absolute error (MAE)					
4-week horizon	0.0319	1.049	1.035	1.051	1.015
13-week horizon	0.0569	0.889	0.882	0.862	0.862
26-week horizon	0.0648	0.679	0.701	0.607	0.672
52-week horizon	0.0783	0.477	0.612	0.382	0.642

Notes: Forecast period is 1990:27 to 1993:52. For the VECM the RMSE or the MAE is expressed in levels. For the alternative forecasts, the RMSE or the MAE is expressed as the inverse of its ratio to the corresponding figure for the VECM. Thus a figure less than 1 indicates superior relative performance by the VECM.

TABLE 7.—RESULTS OF FORECASTING EXERCISES: DOLLAR–MARK

	VECM (level)	VAR (ratio)	Random Walk (ratio)	Forward Premium Regression (ratio)	Forward Rate (ratio)
Root-mean-square error (RMSE)					
4-week horizon	0.0333	0.922	0.954	0.886	0.954
13-week horizon	0.0572	0.728	0.838	0.697	0.837
26-week horizon	0.0605	0.513	0.660	0.438	0.658
52-week horizon	0.0631	0.339	0.637	0.317	0.642
Mean absolute error (MAE)					
4-week horizon	0.0282	1.018	0.993	0.965	0.989
13-week horizon	0.0543	0.893	0.946	0.847	0.944
26-week horizon	0.0636	0.695	0.835	0.775	0.827
52-week horizon	0.0502	0.346	0.513	0.467	0.517

Notes: Forecast period is 1990:27 to 1993:52. For the VECM the RMSE or the MAE is expressed in levels. For the alternative forecasts, the RMSE or the MAE is expressed as the inverse of its ratio to the corresponding figure for the VECM. Thus a figure less than 1 indicates superior relative performance by the VECM.

sample forecasting exercise was conducted. Dynamic out-of-sample forecasts of the spot rate up to 52 weeks ahead were constructed using the VECM for each currency for the period of 1990:27 to 1993:52, with the parameters successively reestimated with each new data point. Tables 6, 7, and 8 give detailed results of the accuracy of these forecasts for the sterling, mark, and yen systems, respectively, using the RMSE and the MAE criteria. The tables also compare the forecasting performance of the VECMs with those of four alternative forecasts—a naive random-walk forecast, the forecast produced by using the appropriate forward rate itself, the forecast generated by an unrestricted fourth-order VAR in the five series, and the forecast produced by a standard regression of the rate of depreciation onto a

TABLE 8.—RESULTS OF FORECASTING EXERCISES: DOLLAR–YEN

	VECM (level)	VAR (ratio)	Random Walk (ratio)	Forward Premium Regression (ratio)	Forward Rate (ratio)
Root-mean-square error (RMSE)					
4-week horizon	0.0282	1.021	0.989	0.991	0.986
13-week horizon	0.0561	0.992	0.965	0.946	0.966
26-week horizon	0.0575	0.679	0.666	0.653	0.667
52-week horizon	0.0622	0.779	0.623	0.458	0.625
Mean absolute error (MAE)					
4-week horizon	0.0213	0.995	0.959	0.957	0.951
13-week horizon	0.0448	0.993	0.943	0.947	0.939
26-week horizon	0.0485	0.786	0.695	0.666	0.688
52-week horizon	0.0512	0.755	0.587	0.445	0.586

Notes: Forecast period is 1990:27 to 1993:52. For the VECM the RMSE or the MAE is expressed in levels. For the alternative forecasts, the RMSE or the MAE is expressed as the inverse of its ratio to the corresponding figure for the VECM. Thus a figure less than 1 indicates superior relative performance by the VECM.

constant and the lagged forward premium.[13] The latter two forecasts were also produced by a process of recursive reestimation.

In each case we report the level of the RMSE and the MAE for the VECM forecasts, but for the alternative forecasts we express the results as the ratio of the RMSE or the MAE from the VECM to that obtained by the alternative method. In table 6, for example, the level of the RMSE of the VECM forecast for the dollar–sterling rate 52 weeks ahead is 0.0828, whereas the ratio of this to the RMSE at the 52-week horizon obtained using a random-walk forecast is 0.575. This indicates a 42.5% reduction in RMSE by using the VECM forecast as opposed to the naive random walk.

The results are broadly similar across the three exchange rates and using either the RMSE or the MAE criterion. At the 4-week horizon, although there is slight evidence of overall superiority of the VECM forecasts, there is little to choose between the five forecasting methods. At higher forecast horizons, however, each of the alternative forecasts is progressively outperformed by the VECM forecast. At the 13-week horizon, the average improvement in RMSE using the VECM forecast is 17.5% for dollar–sterling, 22.5% for dollar–mark, but only 3.3% for dollar–yen. At the 26-week horizon, the corresponding figures are average improvements of 36.8%, 43.3%, and 33.4%, respectively, and at the 52-week horizon, 51%, 51.7%, and 37.9%, respectively.

These results are all the more impressive when one recalls that the forecasts are entirely dynamic, with no extraneous information dated later than the date of the forecast. This contrasts with, for example, model-based forecasting exercises such as Meese and Rogoff (1983), which use information on exogenous variables dated later than the forecast date and are still unable to beat the random walk convincingly.

For dollar–mark and dollar–sterling, the unrestricted VAR fails to outperform the random walk at most horizons,[14] although for the dollar–yen the VAR does in fact outperform the random walk at all horizons, using either criterion. Nevertheless, for all exchange rates and for all horizons beyond the 4-week horizon, the VAR is progressively outperformed by the VECM. While the VAR can be interpreted as an unrestricted form of the VECM, the loss in efficiency from not imposing the error correction and exclusion constraints is clearly important. Similarly, the superiority of the VECM forecasts over the forward premium regression forecasts demonstrates that the standard spot-forward regression fails to extract much of the relevant information from the term structure with respect to future movements in the spot rate.

While we have provided no formal statistical tests for the superiority of the VECM forecasts, the length of the out-of-sample period and the consistent superiority across the range of exchange rates and forecasting horizons considered does make a convincing case for the superiority of the VECM relative to the alternative models.[15]

VII. Conclusion

In this paper we have developed an empirical framework and presented econometric evidence which demonstrates that forward foreign exchange premiums contain significant information regarding subsequent movements in the spot foreign exchange rate. Independently of whether or not foreign exchange markets are characterized by risk aversion or a failure of the rational expectations hypothesis, it appears that the market mechanism is relatively successful in imparting information into the term structure of forward premiums in this respect.

These results raise important issues for further empirical and theoretical research in international finance.

First, in order to establish baseline results, we have restricted ourselves to a linear framework. It is, however, an implication of asset market approaches to the exchange rate

[13] In other words, we project the rate of depreciation $s_t - s_{t-n}$ onto a constant and the lagged forward premium $f_{n,t-n} - s_{t-n}$ using data up to time t, and then form the forecast of s_{t+n} as $[s_t + \alpha_t + \beta_t (f_{n,t} - s_t)]$, where α_t and β_t are the fitted values of the intercept and slope parameters.

[14] That is, the ratios of VECM RMSEs or MAEs to those for the VAR are in most cases smaller than the ratios of VAR RMSEs or MAEs to those of the random walk.

[15] Nevertheless, it should be noted that the smaller number of data points used in computing the statistics for the longer forecast horizons means that their statistical reliability is reduced. This has to be weighed against the increasingly marked apparent superiority of the VECM at the longer horizons.

that risk premiums are likely to be nonlinear (see, for example, Bekaert and Hodrick (1993)). Further empirical work might therefore usefully extend the present analysis to examine nonlinearities.

Second, given that we have established that the term structure of forward premiums contains useful information with respect to future spot rate movements, the question as to precisely why the simple efficient markets hypothesis has been so decisively rejected in the literature acquires fresh urgency. Logically the rejection may be due to deviations from rational expectations or the presence of time-varying risk premiums or, perhaps most likely, both. A third potential alternative is the presence of peso problems, interpreted in the broad sense as small-sample problems in statistical inference. Further investigation of the relative importance of these potential causes of the failure of the simple efficiency hypothesis seems warranted.

The results reported in this paper constitute tentative evidence on a stylized fact concerning the high information content of the forward rate. Further empirical work might be addressed toward establishing the robustness of these conclusions, while further theoretical work should incorporate them.

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